

Module 04: Cognition — Beliefs as Physical States, the Embodied Mind, and Consciousness

Learning Objectives

1. Explain the Active Inference claim that beliefs are physically instantiated states of a generative model, not abstract propositions.
2. Analyze the relationship between precision, attention, and consciousness within the Active Inference framework.
3. Evaluate the "dark room problem" and other philosophical challenges to Active Inference as a theory of cognition.

Introduction

Cognition — thinking, believing, reasoning, attending — is perhaps the most philosophically fraught concept in our entire curriculum. What is a belief? Is it an abstract proposition stored in the mind, a dispositional state of the organism, or a physical configuration of neural activity? Does cognition require consciousness? And how does the brain decide what to think about?

Active Inference answers these questions radically: **beliefs are probability distributions over the hidden states of the world, physically realized in the internal states of the agent.**

Cognition is not a separate faculty "above" perception and action but the same inferential process operating at different timescales and levels of abstraction. This module explores what this implies philosophically.

Key Concepts

1. Beliefs as Physical States

In the Active Inference framework, a "belief" is not a sentence in a language of thought (Fodor) nor a disposition to behave (Ryle). It is a **posterior probability distribution** $Q(s)$

over hidden states, physically realized in the brain's internal states. These beliefs are updated continuously as new observations arrive — cognition is ongoing inference.

This places Active Inference in the tradition of **physicalism** about mental states, but with an important twist: the physical states are not arbitrary neural firings but states that are *about* something — they carry representational content because they are components of a generative model that maps onto the causal structure of the world.

2. Precision, Attention, and the Architecture of Cognition

Not all beliefs are created equal. Some predictions are held with high confidence (high **precision**), others with low confidence. Precision, in Active Inference, plays the role of attention: precision-weighting determines which prediction errors are amplified and which are suppressed.

This accounts for a range of cognitive phenomena:

- **Selective attention:** High precision on one sensory channel suppresses others
- **Cognitive flexibility:** Adjusting precision allows switching between habitual and deliberate modes of thought
- **Psychopathology:** Aberrant precision (too high or too low) may underlie conditions like anxiety (overweighting threat predictions) and autism (overweighting sensory prediction errors)

3. The Dark Room Problem and Active Cognition

The "dark room problem" is a standard objection to Active Inference: If agents minimize surprise, why don't they simply find a dark, quiet room and stay there — minimizing all sensory input? The answer reveals something deep about cognition: agents minimize *expected* surprise relative to their **prior preferences** (the C vector). A human has priors for social contact, nutrition, warmth, and curiosity. Staying in a dark room would *violate* these priors and thus *increase* expected free energy.

Philosophically, this connects to the Aristotelian idea that living well requires the active exercise of capacities, not mere passive comfort.

Applications

- **Consciousness:** If cognition is inference, is consciousness the *experience* of inference? Or can inference proceed unconsciously? This connects to the "hard problem" of consciousness (Chalmers, 1995).
- **Extended Cognition:** If beliefs are physical states of a generative model, can external artifacts (notebooks, smartphones) be part of the model? This returns us to Clark and Chalmers' extended mind hypothesis.

Conclusion

Cognition, in Active Inference, is not a mysterious inner theater but the ongoing process of belief updating under precision-weighted prediction error. This view has radical implications for consciousness, psychopathology, and the boundaries of the mind. Module 05 examines how beliefs are translated into action.